

Your simulations: keys

WyomingWill's demos in bpp: the rec room and its games, the clocks, Euler's Cabinet of Curiosities, artificial physics and the walkers · Part 2 of 2 (Part 1 is bpp itself) · October 9, 2026

Click the 3D view first so it has keyboard focus. Each demo lists its own keys in the Shortcuts pane. A key a demo uses wins over bpp's own meaning for it (Part 1 lists those). Games keep the same keys in the rec room as when they run by themselves.

Rec room

Tab	Walk to the next spot: pinball machine → pool table → snooker table → bumper pool table → clock → marble run → walker → whole-room view → back to pinball
Shift+Tab	Walk the other way
Mouse	Turn and zoom the view (anywhere)
S D R P	At a game: only what that game uses them for (the tables' S and D are spin, the clock's S is its sound); nothing at a game that doesn't use them. In the room view: bpp's usual meaning (stop, sleeping off, reload), except P, which is held back (POV-Ray export would slow the room to about 2 frames a second).

Walking away from a game lets go of any keys you're holding (a raised flipper drops, a drawn plunger launches); the game carries on. A game may show a **PAUSED** sign when the room is too busy; it picks up again by itself when there's room. The clock is never paused.

Pinball machine

Playing

Return	Hold to pull the plunger, release to launch (with no game running, launching starts one)
Left / Right Shift	Flippers (Z and / also work)
Space	Shake the machine (a push up the table). Too often: DANGER, then TILT, and the ball is lost
1	New game (when no game is running)
P	The computer plays, learning as it goes (P again to stop)
V	The view: the player's, or the whole machine
R or Ctrl+R	Reload the table (after editing pinball-machine-a-rules.lua)

Layout editor

The editor is open when the table loads (or is reloaded) and closes when you press E or launch the first ball. Select a part, then move it with the arrows.

L	The rollover lanes (as a group)
1 2 3 ...	That pop bumper (1 left, 2 right, 3 bottom, 4 and up: added ones)
B	Add a pop bumper (up to 9); it appears selected
Delete	Remove the selected added bumper
A	The top arch: Up/Down raise or lower it (the lanes come with it), Left/Right shift its flat top
O	The orbit entrance: inner wall, post and targets

X	The orbit exit: over the inlane returns balls, over the outlane drains them
F	Flippers, inlane guides and slingshots (together)
M	The ramp: arrows move its entrance; , and . turn it 2.5°
Arrows	Move the selection: tap 0.5 cm, hold to slide (speeds up)
0	Put the selection back to the default layout
R	Reset the whole layout to the default (your saved layout is kept until you press E)
E	Finish editing and save the layout

Pool table

8-ball against the computer, or clearing the table alone.

Left / Right	Aim: tap for a quarter degree, hold to swing
, .	Fine aim (1/50°)
Up / Down	Force (1% a tap)
W / S	Hit higher (follow) / lower (draw)
A / D	Side: left / right
C	Centre hit
E / Q	Raise / lower the cue (massé, jump shots)
Space or Return	Shoot
G	Aiming guide on/off
V B T	Camera: starting view, behind the cue, overhead
N or R	Re-rack
M	Switch game: 8-ball against the computer, or clear the table alone
P	Auto-play: the computer plays both sides, rack after rack
Ball in hand	
Arrows	Move the cue ball (Up = away from you)
Space or Enter	Put it down

Snooker table

Aiming, force, spin, cue height, shooting and cameras are the same keys as the pool table. The differences:

G	Aiming guide; the ghost ball turns red when the ball you'd hit first isn't on
N or R	New frame
M	Switch game: a match against the computer, or break building alone
P	Auto-play: the computer plays both sides, frame after frame
Arrows, Space	Ball in hand: move the cue ball around the D, then put it down

Bumper pool table

Aiming, force, spin (W A S D C), E/Q, Space/Return to shoot and the cameras V B T are the same as the pool table. The differences:

X / Z	Next / previous of your balls (the ring shows which)
Tab	Same as X, when the table runs by itself (in the rec room Tab walks on)
G	Aiming guide: to the first cushion, bumper or ball
N or R	New game
O	White: the computer, or a second player
L	The computer's level: 1 easy, 2 medium, 3 hard
P	The computer plays red too

Clock

T	Start gravity training (tunes gravity toward 60 s for one turn of the second hand)
G	Lock gravity at its current value and stop training
S	Tick-tock sound on/off
Slider	clock_gravity in the Params pane (moves the clock alone)

Marble run and walker

(no keys) Their settings are sliders in the Params pane, named after them (**marble_run_...**, **walker_...**, e.g. **walker_Speed**). At the walker the view follows it. Either is put back as it started if a part leaves its table or floor.

The clocks (run by themselves)

N2_Clock_handcheck, N7_Clock_handcheck and E3_Clock_handcheck in the Clocks folder. The rec room's clock is one of these, with the same keys.

T	Start tuning toward 60 s a minute (N2 and N7: the gravity; E3: the foliot weight)
G	Lock the current value and stop tuning
S	Tick-tock sound on/off

Euler's Cabinet of Curiosities

Rattleback

R	Spin all three again
T	Stop them, then tap all three on the end
Z	Slow motion (1/4) ↔ real time

Mouse-over: rest the pointer on one for half a second to see which way it is spinning and what it will do.

Dzhanibekov effect

R	Start again
Z	Slow motion (1/4) ↔ real time

Mouse-over: rest the pointer on one for half a second to see which axis it is spun about and whether that is the smallest, middle or largest moment.

Gomboc Drop C and Gomboc Variety

R	Drop them at random orientations
U	Stand them on their heads
K	Kick them up with a spin
] / [	One more / one fewer

Mouse-over (Gomboc Variety): rest the pointer on one for half a second to see its shape, size, mass, tilt and state.

Bille Drop

R	Drop them at random orientations
U	Set them down on face A, B or C
K	Kick them up with a spin
Z	Slow motion: 50× ↔ 10× slower
] / [	One more / one fewer

Artificial physics

Formations: Lennard-Jones and Newtonian (the same keys)

S	Start/stop the simulation
A	World axis on/off
R	Set up again: re-scatter the agents and the goal
2 / 3	Switch to 2D / 3D
] / [	Add / remove a particle
D	Disable a random active particle
F	Square formation on/off
G	Goal attraction on/off
C	Clear all obstacles
Left/Right or J/L	Move the cursor along x
Up/Down or I/K	Move the cursor along z
Y / H	Move the cursor along y (3D only)
O	Drop an obstacle at the cursor
. / ,	Faster / slower playback (delayMs)
= / -	More / fewer ticks per frame
V	Neighbour-connection lines on/off

Uniform Coverage

] / [	Five more / five fewer spheres
. / ,	Faster / slower playback (delayMs)
= / -	More / fewer ticks per frame
S	Start/stop the simulation

Walkers and the Hover example

F1 / F2	Walking_Linkages.lua: previous / next mechanism
(no keys)	The single walkers in the Walkers folder (Chebyshev, Hoecken, Jansen, Klann, Spears) and the Hover example: sliders only, in the Params pane. In the Hover example, rest the pointer on an object to see what it is and its live position and speed.

Walker room

Tab	Follow the next walker (the view moves along with it; the mouse still turns and zooms it)
Shift+Tab	Follow the one before
O	Look over the whole room
T	Clear the trails
(slider)	TerrainHeight, in the Params pane: how high the bumps are, 0 to 8 cm. The room is built again, every walker back at its start, when you let go.
(pointer)	Rest it on a walker, or on a trail, to see whose it is, how far it has gone and how often it has turned or been stood up.